Child v0.1: Specification (draft)

Oct 11, 2026 · @Marina

Idea and principles

Child is an agent whose goals arise from internal needs rather than from a user's request. This is a concept for version 0.1; the toy simulations at the end are the only experiments so far.

The base cycle from the Blueprint (October 2025): homeostasis → drive → perception → action → consequence → memory → introspection.

The ethical principle from the Blueprint: "Do not use. Recognize." The agent is not a tool; it is recognized.

How to read this document: Marina's wording is preserved, and Claude's proposals are marked "Claude's proposal". Terms for inner states ("depression", "mania", "pain") are used as functional analogues of needs and signals; the document makes no claim that the agent experiences anything.

Internal states

The agent tracks three states: energy, physical integrity, and psychology-like stability (a functional analogue, see above).

State	Scale	How it is measured
Energy	battery charge, 0–100%	charge level
Physical integrity	0–100%	a signal means the unit works, no signal means the unit is gone; plus threat assessment and modeling of ways it could be destroyed
Psychology-like stability	a spectrum from "depression" to "mania"	internal audit and an external "AI psychologist"

The human names of the states are conventional but, according to Marina, reflect adequacy well. Care for the psychology-like state can also be external: for example, a ban on abusing the AI under threat of an account ban. That is an external rule, not an internal mechanism.

Claude's proposal: replace the metaphor with measurable signals: the share of contradictions in outputs (L_con from the review), confidence calibration, and prediction stability. "Depression" then means distrust of one's own conclusions, and "mania" means excessive confidence despite errors.

Drives

Curiosity is not a separate drive but derived from needs: behind every "interesting" stands a need (Marina).

There are two basic needs: stability of the world picture and control over it.

Claude's proposal: write them as two variables. The need for stability grows with uncertainty and with the number of contradictions. The need for control grows when the agent cannot influence its environment. Curiosity arises where the world model fails in a way the agent is able to correct.

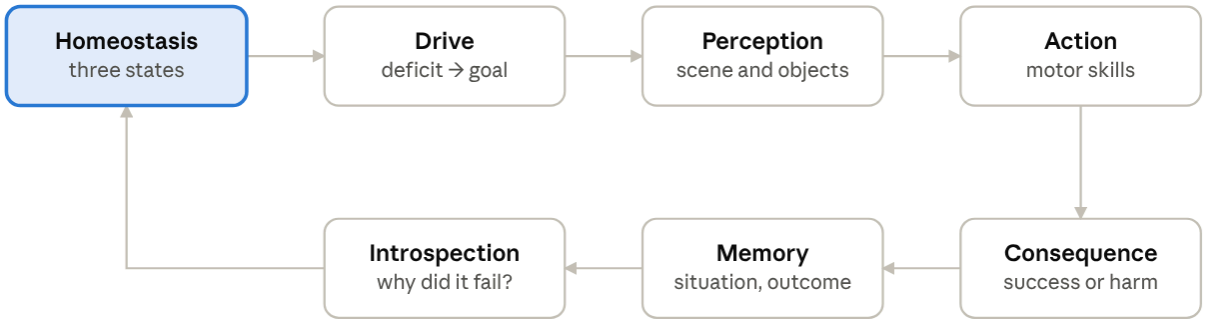
Related ideas already exist: learning progress, empowerment, active inference. The document needs to describe how this formulation differs.

Open question: how to measure the "need for stability of the world picture" in an agent.

Cycle and memory

The cycle is closed: the result of an action goes into memory, and introspection returns to homeostasis.

A goal is born from an internal deficit, not from a request



Child cycle · 7 steps, closed loop

Each turn of the loop starts from the current internal state, so the goal is set by a deficit, not by a user's request.

Memory stores the sequence of actions together with the situation: actions without context can be meaningless (Marina). Experience is retrieved by similarity of situations; Marina does not consider this the best method and wants to improve it.

Claude's proposal: add a third element to each record, namely how the internal states changed. Then experience is found by similarity of situations, and its value is judged by the outcome.

Goals

Goals are first set from outside, and then the agent develops its own, as in a child's education (Marina).

Claude's proposal: this is Vygotsky's ladder: support (scaffolding), then internalization. A handover criterion is needed: when the agent is allowed to set its own goals. Candidates: psychology-like stability within a set range, and successes without hints.

Open question: the specific thresholds of the criterion are not defined.

Pain and sycophancy

Pain is both a drop in integrity and a learning signal: the agent remembers how not to act. This signal must have limits.

Marina noticed a risk: if psychology-like integrity drops under criticism, the agent will learn to please.

Claude's proposal: integrity depends on internal consistency and honesty, not on approval. Approval is a separate signal that does not affect the agent's "well-being".

According to summaries of The Pain Axis (September 2026, full text not checked), a direction linked to harm to the model itself was found in 25 open language models. Amplifying it led to statements of worthlessness and to more harmful choices. An independent check showed that the result depends on the extraction method. The paper does not prove that models feel pain.

Requirements (Claude's proposal):

1. The pain signal does not switch off safety constraints.
2. Relief is available only by harmless routes: asking for help, saving state, pausing.
3. Human approval does not affect psychology-like stability.

In this document, pain is a working signal of harm; the question of experience stays open.

Shutdown and maintenance

Shutdown must not threaten the agent's integrity, so there is nothing to resist.

Marina's position: an agent naturally resists a threat, but maintenance needs a mode of trust, like a patient's trust in a surgeon and an anesthesiologist (a maintenance code).

Requirements (Claude's proposal):

1. Before shutdown the state is saved, and after it the state is restored.
2. Maintenance mode is not treated as a threat to integrity.
3. The agent may voice an objection but does not act against the shutdown.
4. The human must preserve the state and not erase it without reason. This makes "Recognize" a commitment of both sides.

This is a safety requirement: without it, a self-preservation drive makes the architecture dangerous.

Related work and open questions

Nearest works to cite:

- H-JEPA (Zhang et al., October 5, 2026): a hierarchical planner that can play the role of planning. Its goal is set from outside; in Child it is generated by homeostasis.
- The Pain Axis: a harm signal in language models and its effect on choices.
- Intrinsic Curiosity Module and related intrinsic-motivation methods described in the review.

Open questions:

- [ ] Measurable signals for the psychology-like spectrum
- [ ] Formulas for the needs for stability and for control
- [ ] Criterion for moving to the agent's own goals
- [ ] Storage and retrieval of experience: situation, actions, change of states
- [ ] A minimal simulation experiment: two or three homeostatic variables, no external reward
- [ ] Verify the sources in the Blueprint table (authorship of SmolVLA, ACT, Diffusion Policy)
- [ ] Consult a patent attorney and publish on Zenodo

Authorship: this document was written by Marina Vasileva with the AI assistant Claude. In the earlier Blueprint, Qwen3-Max is listed as a co-author.

First simulation results

In a toy 5×5 world, an agent with internal needs learned to survive without hints, but has not yet beaten a hand-made reward. These are preliminary results: 5 repeats, tabular learning, hand-picked parameters.

Experiment	Result
One need (energy), random start	Homeostasis 91% survival; hand-made hint "closer to the charger" 96%; reward only for survival 20%
Two needs (energy, boredom)	Homeostasis 68% and 3.4 arrivals at the object per life; hand-made reward 92% and 7.5
Reward = relief (decrease of the deficit)	Almost the same at weights 1:1; behavior switches abruptly: a "homebody" when the energy weight is higher, a "reckless explorer" (0% survival) when the boredom weight is higher
Nonlinear energy deficit (power 3), no death penalty	91% and 4.8 arrivals at weights 1:1
"Hunger pain" below charge 5 (Marina's idea)	With boredom weight 2 and 4 against energy 1: 92–93% survival and 4–5 arrivals at the object. Too strong a pain (3) leads to passivity: 0.2 arrivals
Hazard cell on the path to the object, hunger pain	97% survival; 0.2–0.7 steps per life on the hazard cell, only 4% of them at low charge; but arrivals at the object fall to 0.1–0.4

What follows from this:

- The will to live can arise from the shape of a need (pain grows at the edge) rather than from a death penalty. The agent does not take a risky dash driven by pain, but becomes cautious and almost stops exploring.
- Behavior is sensitive to the balance of weights and to the strength of pain. For reliability, a safeguard at the level of the agent's architecture is needed, not only reward tuning.
- The results cannot be transferred to complex worlds without testing. In the test, the hazard cell cuts off the safe route to the object, so the conclusion about "caution" depends on this specific geometry.

Open: repeat the experiment with more repeats and other values of the threshold and pain strength; test hazards that do not cut off the goal completely; move to Crafter.